Smart Expense Tracker

Smart Expense Tracker is a real Python command-line project that analyzes personal finance transactions from a CSV file. It imports transactions, categorizes common expenses, calculates financial summaries, detects unusually large expenses, and generates a clean Markdown report.

This project is designed to be GitHub-ready. It includes a modern `src` layout, tests, packaging metadata, a sample dataset, and continuous integration configuration.

Project Overview

Area	Details
Language	Python
Interface	Command-line application
Main use case	Analyze income, expenses, category spending, monthly cash flow, and unusual expenses
Data input	CSV file with `date`, `description`, `amount`, and optional `category` columns
Privacy approach	Runs locally and does not upload financial data
Author	Manus AI

Features

The application provides a practical workflow for reviewing personal finance data. It can automatically categorize transactions with keyword rules, show a summary of income and spending, group expenses by category, calculate monthly cash flow, and create a Markdown report that can be committed to GitHub or shared privately.

Command	Purpose
`summary`	Prints transaction count, income, expenses, net cash flow, and savings rate.
`categories`	Prints spending totals grouped by category.

`monthly`	Prints net cash flow grouped by month.
`unusual`	Detects unusually large expenses compared with category averages.
`report`	Generates a Markdown report from the CSV file.

Installation

Clone the repository after you upload it to GitHub, then install it in editable mode.

Bash

```bash
git clone https://github.com/YOUR_USERNAME/smart-expense-tracker.git
cd smart-expense-tracker
python -m venv .venv
source .venv/bin/activate
pip install -e ".[dev]"
```

CSV Format

The CSV file must include `date`, `description`, and `amount`. The `category` column is optional. Positive amounts are treated as income, while negative amounts are treated as expenses.

Plain Text

```
date,description,amount,category
2026-01-01,Payroll Deposit,3200.00,
2026-01-02,Apartment Rent,-1250.00,
2026-01-04,Aldi Grocery Store,-82.40,
```

Usage

Run the commands below from the project root after installation.

Bash

```bash
smart-expense-tracker summary sample_transactions.csv
smart-expense-tracker categories sample_transactions.csv
smart-expense-tracker monthly sample_transactions.csv
```

```
smart-expense-tracker unusual sample_transactions.csv
smart-expense-tracker report sample_transactions.csv --output expense_report.md
```

You can also run the package directly without installing the command-line script.

Bash

```bash
python -m smart_expense_tracker.cli summary sample_transactions.csv
```

Example Output

Plain Text

```
Transactions: 15
Income: $6,400.00
Expenses: $3,256.44
Net cash flow: $3,143.56
Savings rate: 49.12%
```

Development

The project includes automated tests written with `pytest` and linting support with `ruff`.

Bash

```bash
pip install -e ".[dev]"
pytest
ruff check .
```

Project Structure

Plain Text

```
smart-expense-tracker/
├── .github/workflows/ci.yml
├── src/smart_expense_tracker/
│   ├── __init__.py
│   ├── analyzer.py
│   ├── categorizer.py
│   ├── cli.py
│   ├── importer.py
│   ├── models.py
```

```
│   └── report.py
├── tests/
│   ├── test_analyzer.py
│   └── test_importer.py
├── sample_transactions.csv
├── pyproject.toml
├── README.md
├── LICENSE
└── .gitignore
```

GitHub Upload Instructions

Create an empty repository on GitHub named `smart-expense-tracker`, then run these commands inside the project folder.

Bash

```bash
git init
git add .
git commit -m "Initial commit: add smart expense tracker"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/smart-expense-tracker.gi
git push -u origin main
```

If you already use the GitHub CLI and are logged in, you can create and push the repository with this command.

Bash

```bash
gh repo create smart-expense-tracker --public --source=. --remote=origin --push
```

License

This project is released under the MIT License.